AICTE – BharatCares – IBM SkillsBuild Machine Learning and Applied AI Internship 2026

PROJECT CONCEPT NOTE & LEAN CANVAS

Capstone Project Final Submission | Masterclass 5 Framework

Project Title:	PHISHGUARD//NEXUS: Explainable AI Phishing & Scam Detection Platform
Student Name:	Sahil Rai | **Degree Program:** Bachelor of Computer Applications (BCA)
Internship ID:	IBMUEML002432 | **Institution:** Lovely Professional University (LPU), Punjab
Live Cloud URL:	https://phishguard-nexus.vercel.app/ (Shortlink: tinyurl.com/pg-nexus)
GitHub Repository:	https://github.com/rootbysahil/phishguard-nexus

1. Executive Summary & UN Sustainable Development Goals (SDGs)

PHISHGUARD//NEXUS is an Explainable AI (XAI) cybersecurity system engineered to detect zero-day phishing across URLs, emails, and SMS without external network lookups. Operating on 24 pure-Python lexical signals and a calibrated Logistic Regression classifier (ROC-AUC: 1.000, F2: 0.9989), it integrates directional SHAP feature attributions, IBM watsonx / Granite LLM multilingual advisories, and direct integration with India's National Cybercrime infrastructure (Helpline 1930 / cybercrime.gov.in).

- **UN SDG 9 (Target 9.c):** Equitable, zero-latency phishing defense infrastructure accessible to Indian citizens, MSMEs, and educational bodies.
- **UN SDG 16 (Target 16.4):** Actively counters financial cybercrime, identity harvesting, and digital extortion.

2. Problem Statement & Critical Industry Inefficiencies

1. **Lagged Static Blacklists:** 70%+ of phishing attacks expire within 4 hours; static feeds like Google Safe Browsing lag by 12–48 hours.
2. **Black-Box Neural Networks:** Deep learning models output raw probabilities without actionable forensic explanations, causing user distrust.
3. **Privacy Leakage & Latency:** Standard scanners perform live DNS/HTTP queries, alerting attackers and exposing victim IP addresses.
4. **Lack of Vernacular Recourse:** Non-technical citizens lack guidance in regional languages (Hindi, Punjabi) or immediate emergency escalation.

3. The Lean Canvas (9-Box Business & Technical Framework)

<table>
<tr>
<td rowspan="2">1. PROBLEM
• Zero-day links bypass blacklists (<4h lifespan).
• Black-box AI causes mistrust.
• Tools leak victim IP via live lookups.

Existing Alternatives:
VirusTotal, Google Safe Browsing.</td>
<td>4. SOLUTION
• 24 pure-Python lexical features (0 network calls).
• Calibrated ML (ROC-AUC 1.000, F2 0.9989).
• Directional SHAP XAI.
• Multilingual watsonx advisories.
• Automated forensic PDF reports.</td>
<td rowspan="2">3. UNIQUE VALUE PROP
Core Message:
"Forensic, zero-latency defense that explains WHY a link is dangerous and WHAT TO DO in regional languages."

High-Level Concept:
VirusTotal + IBM XAI + 1930 Helpline.</td>
<td>9. UNFAIR ADVANTAGE
• Zero-leakage domain-grouped training split.
• Privacy-first zero-call architecture.
• Native Indian Cybercrime 1930 / portal escalation.</td>
<td rowspan="2">2. CUSTOMER SEGMENTS
• Everyday Indian mobile & UPI banking users.
• Elderly citizens prone to KYC scams.
• MSMEs without dedicated SOCs.

Early Adopters:
University students & fintech users.</td>
</tr>
<tr>
<td>8. KEY METRICS
• ROC-AUC (1.000) & F2 (0.9989)
• Latency (<15ms)
• Citizen scans logged & PDF audits generated</td>
<td>5. CHANNELS
• Vercel Web App (tinyurl.com/pg-nexus)
• Mobile QR Code & PWA
• AICTE & IBM SkillsBuild Showcase</td>
</tr>
<tr>
<td colspan="2">7. COST STRUCTURE
• Vercel Serverless Hosting ($0 Hobby)
• Model Compute: Local / Edge SQLite
• IBM watsonx.ai Token Usage</td>
<td colspan="3">6. REVENUE / SUSTAINABILITY STREAMS
• Free Community Tier: Public citizen protection
• Enterprise B2B API: Pay-per-scan endpoint for Indian fintechs & banks
• Institutional Audits: Branded forensic compliance reports</td>
</tr>
</table>

4. Machine Learning & Applied AI Architecture

• **Zero-Network Feature Extraction:** Evaluates 24 lexical, syntactic, and structural signals (Shannon entropy, Punycode/homoglyph detection, brand keyword displacement in subdomains, path depth, TLD risk analysis) with 0 network calls, ensuring zero latency and zero victim IP exposure.
• **Champion Classifier:** Calibrated Logistic Regression using Platt probability scaling, mitigating path-length bias with balanced benign samples.
• **Validation Rigor:** Strict GroupShuffleSplit on registered domains; zero test-train domain leakage.
• **Explainability (XAI):** LinearExplainer assigning directional weights to 24 features (entropy, brand displacement, homoglyphs).
• **IBM watsonx.ai Integration:** Synthesizes forensic findings into plain language and powers regional translations (English, Hindi, Punjabi).
• **Automated Forensic PDF:** Real-time audit generation stamped with SHA-256 cryptographic verification.

5. IBM SkillsBuild 6-Week Masterclass Milestones Alignment

Week	SkillsBuild Masterclass	Project Accomplishment Executed
Week 1	Kickstart ML & Tools (VS Code, BOB)	Environment setup, dataset collation & API contracts.
Week 2	Supervised Learning & IBM watsonx	24-feature extractor, baseline heuristics & classifier benchmarking.
Week 3	Unsupervised Patterns & Deep Learning	Shannon entropy, homoglyph clustering & GroupShuffle validation.
Week 4	Intelligent Agents & Lifelong Skills	SHAP XAI engine, watsonx advisory prompt orchestration.
Week 5	Project Completion & Q&A	FastAPI backend, Cyberpunk UI & ReportLab PDF engine.
Week 6	Project Submission (PPT & Code)	Live Vercel cloud deployment, automated test verification (8/8).

6. Performance Evaluation & Champion Model Benchmark

Metric	Logistic Regression (Champion)	Random Forest	Decision Tree
ROC-AUC Score	**1.0000**	0.9995	0.9850
F2 Score (Recall-Weighted)	**0.9989** [0.997, 1.000]	0.9940	0.9810
Recall (Phishing)	**1.0000** (0 Missed)	0.9950	0.9820
Inference Latency	**<15ms**	~45ms	<20ms
Explainability	**Exact SHAP Linear Coefficients**	Tree approximations	Opaque branches

7. Student Undertaking & Submission Declaration

I hereby declare that this project, **PHISHGUARD//NEXUS**, has been developed as the final capstone deliverable for the 6-week virtual IBM SkillsBuild Machine Learning and Applied AI Internship (2026) in collaboration with AICTE and BharatCares. The platform is fully operational, publicly hosted at **https://phishguard-nexus.vercel.app/** (accessible via **tinyurl.com/pg-nexus**), and backed by a clean version-controlled repository on GitHub.

Student Name: Sahil Rai | **Degree:** Bachelor of Computer Applications (BCA)
Internship ID: IBMUEML002432 | **Institution:** Lovely Professional University (LPU), Punjab
Student Signature: Sahil Rai **Date of Submission:** 13/10/2026